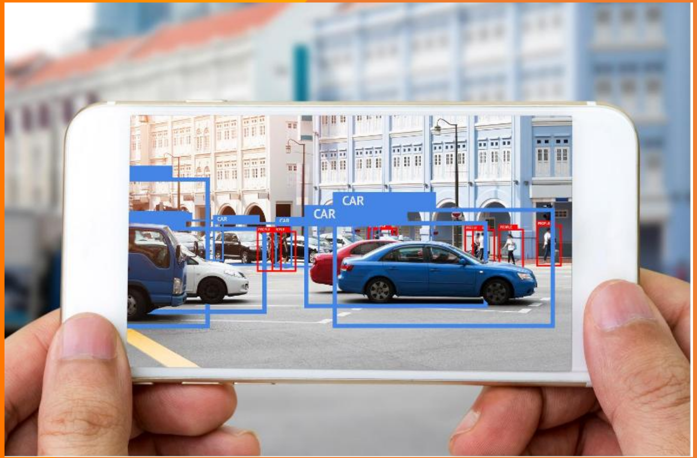
Module 03: Deep Learning Applications: From ChatGPT to Computer Vision



Courtesy to Andrew Ng Machine Learning and Deep Learning Course DeepLearning.AI

Created by Head Digital Learning Advisor

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Learning Outcomes

- By the end of this workshop students should be able to:
- Explain major application areas of Deep Learning.
- Compare NLP, Speech Recognition, and Computer Vision applications.
- Identify datasets, models, and challenges associated with different applications.
- Critically evaluate the opportunities and risks of Deep Learning.
- Develop an application idea relevant to Assessment 1.

Module-02 Recap

- Feedforward Networks
- Forward Propagation
- Loss Function
- Backpropagation
- Weight Updates

Recap Quiz

Question 1: What is the primary characteristic of a Feedforward Neural Network?

- A. Information flows in both directions between layers.
- B. Information only flows from the output layer to the input layer.
- C. Information flows from the input layer through hidden layers to the output layer.
- D. The network does not contain weights.

Answer: C

- **Discussion:** Why is it called a *feedforward* network?

Question 2: What happens during Forward Propagation?

- A. The network updates its weights.
- B. The network calculates gradients.
- C. The network passes input data through the layers to generate a prediction.
- D. The network propagates errors backwards.

Answer: C

Recap

Question 3: What is the purpose of a Loss Function?

- A. To increase the number of neurons in the network.
- B. To measure how different the predicted output is from the actual output.
- C. To remove hidden layers.
- D. To store training data.

Answer: B

- **Discussion:** Why does a neural network need feedback during learning?

- **Question 4:** What is the main purpose of Backpropagation?
- A. To generate training data.
- B. To transfer inputs from one layer to another.
- C. To distribute error information backward through the network and determine how weights should be adjusted.
- D. To add new hidden layers automatically.

Answer: C

- **Discussion:** Why can't only the output neuron learn from the error?

Recap

Question 5 – When should a neural network update its weights?

- A. Before making a prediction.
- B. After calculating the error and determining how much each weight contributed to that error.
- C. Only when the prediction is correct.
- D. Randomly during training.

Answer: B

- **Discussion:** What would happen if the network never updated its weights?

Reflection Question (Think–Pair–Share)

- A neural network predicts that a student will score **40**, but the actual score is **80**.
- Explain, in your own words, the sequence of steps that occur from prediction to learning.

Assessment Progression

- **Assessment 1 is due at the end of Module 4**. Please refer to the assessment briefs in the assessment area for more detail.
- You are expected to spend approximately one hour reviewing Assessment 1 in this Module. You should:
  - Review and familiarize yourself with Assessment 1;
  - Enhance your understanding of the key concepts of deep learning; and
  - Review algorithms, such as N-Gram, in preparation for Assessment 1.
- ***You can prepare for this Assessment by using the learning resources of this Module***, which include readings, videos and learning activities. These resources will enhance your understanding of the key concepts in this Module.

Why This Matters

- ChatGPT
- Siri
- Alexa
- Tesla
- Face Recognition
- Netflix
- Spotify

Discussion:

- Which of these technologies have you used this week?

Artificial Intelligence Applications

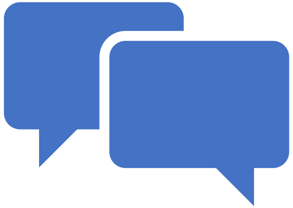
Natural language processing (NLP): A branch of AI that helps machines understand, interpret and manipulate human language;

Speech recognition: Technology employed by smart systems, such as Amazon's Alexa or voice-recognition texting; and

Computer vision: An analytical solution that can achieve state-of-the-art results in image classification, object detection and face recognition.

Additionally, AI and deep learning are redesigning healthcare by bringing in a new era of precision medicine, changing the eCommerce landscape to enable companies to better engage with their customers and improving manufacturing processes across the supply chain.

Recap Cycle 1: Think-Pair-Share



If Deep Learning is so powerful, why isn't it used for every problem?

Discuss. which industry will be transformed most by AI in the next 10 years?

Applications: What is NLP?

- Which NLP application have you used recently?
- Is ChatGPT an application of NLP?

https://www.youtube.com/watch?v=CMrHM8a3hqw

NLP Applications

- Translation
- Chatbots
- Sentiment Analysis
- Text Summarisation

- Any specific Examples you can share?
    - Question/Answering systems,
    - Automated Customer Service System,
    - Tutoring System

Assessment 1 Connection

- Twitter Sentiment Analysis
- You can review the research paper provided in MyLearn for this assessment task or check it later at your convenience.

Questions:

- What is the input?
- What is the output?
- What labels are required?

N-Gram Approach in Natural Language Processing

https://www.youtube.com/watch?v=E_mN90TYnlg

Assessment-1 Resources: Twitter Sentiment Analysis

- **Twitter Sentiment Analysis Using Naive Bayes and N-Gram**
  - Analyzing the level of positivity of tweets
- https://betterprogramming.pub/twitter-sentiment-analysis-using-naive-bayes-and-n-gram-5df42ae4bfc6

- **Twitter Sentiment Analysis- A NLP Use-Case for Beginners**
- https://www.analyticsvidhya.com/blog/2021/06/twitter-sentiment-analysis-a-nlp-use-case-for-beginners/
- Activity – Go through this blog and work in Python.

Assessment-1 Resources : Twitter Sentiment Analysis Using Python

https://www.youtube.com/watch?v=ujId4ipkBio

Assessment-1 Resources: Accessing the Tweets Directly

https://www.youtube.com/watch?v=27P268Q7pE0

Group Activity: Design an NLP Application

Examples:

- Mental Health Monitoring
- Student Feedback Analysis
- Fake News Detection
- Customer Support Chatbot

You can identify:

- dataset
- input
- output
- possible model

Recap Cycle 2: Concept Mapping Activity

Text
↓
Dataset
↓
Model
↓
Prediction
↓
Application

Speech Recognition: Applications

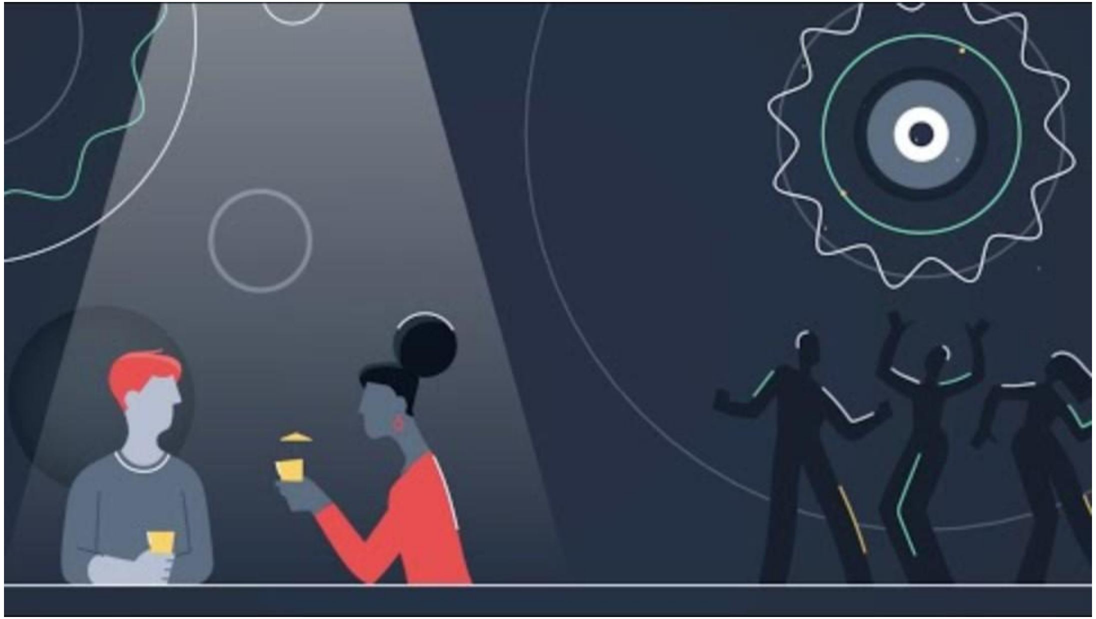
- How Siri understands you?
- Why do we want to predict a words, given some preceding words?
- Applications:
    - spelling correction,
        - Rank the likelihood of sequences containing various alternative hypotheses
            - Theatre owners say popcorn/unicorn sales have doubled...
    - handwriting recognition
    - statistical machine translation
        - Assess the likelihood/goodness of a sentence, e.g. for text generation or machine translation
            - The doctor recommended a cat scan.
    - Others…

Think-Pair-Share

- What challenges make speech recognition difficult?

Example responses:

- accents
- noise
- language differences
- speed



https://www.youtube.com/watch?v=uSNUmJffK4c

Group Design Activity

Design a Speech Recognition System

Examples:

- Healthcare assistant
- Classroom assistant
- Accessibility tool

Recap Cycle 3

Reflection

- Would you trust an AI system to transcribe medical conversations?
- Why or why not?

Computer Vision

- What is Computer Vision?

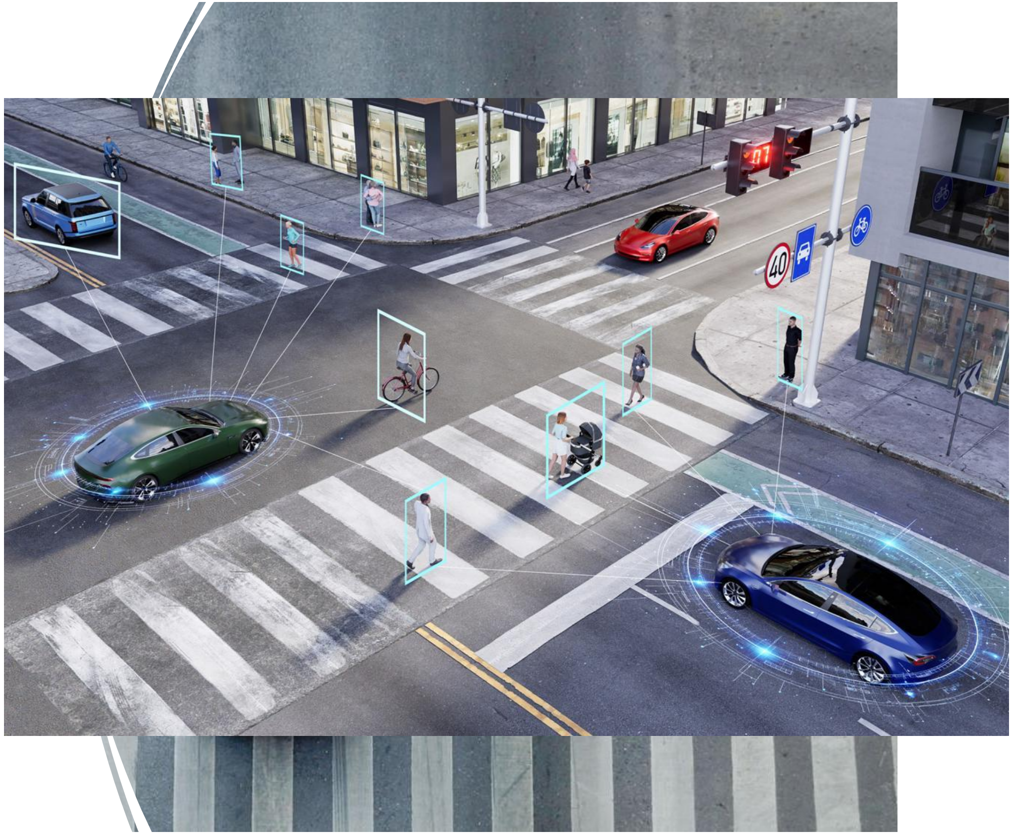
Examples:

- Face Recognition
- Medical Imaging
- Self-Driving Cars
- Agriculture

Image Analysis Activity

- What can humans see?
- What must a computer learn to see? Like: Edges, Shapes, Colours,

Humans Recognise	Computers Must Learn
Objects	Features
Faces	Patterns
Context	Pixel relationships
Meaning	Mathematical representations
Danger	Decision boundaries



Computer Vision Applications

- Autonomous Cars
- Agriculture
  - Crop monitoring
  - Disease detection, Insect Detection
  - Harvesting,
  - Drone detail images for condition of the soil, and crop
- Security Surveillance System
- Healthcare
- Waste Management and Recycling

Autonomous Models

Group Activity: Design a Computer Vision Application

Your task is to identify:

- problem
- input image
- output
- dataset
- ethical concerns

Recap Cycle 4: Compare the Three Domains and complete the table.

NLP	Speech	Vision
Input	?	?
Output	?	?
Challenges	?	?

Human-Centred AI: Benefits and Risks

Benefits:
- productivity
- accessibility
- automation

Risks:
- bias
- privacy
- misinformation
- surveillance

Human Judgement vs AI

Discussion:

- Just because AI can do something, should it?

GenAI Reflection Activity

Ask ChatGPT: What are the risks of Deep Learning?

Compare and critique your own response/understanding with the ChatGPT response.

Key Takeaways

- Deep Learning powers NLP, Speech, and Vision.
- Different applications require different models.
- Human judgement remains critical.
- Assessment 1 should now feel more concrete.

Learning Activity-01: NLP/Speech Recognition/Computer Vision

- Step 1: Think of an exciting **NLP/Speech Recognition/Computer Vision** application that is important to you and that you believe that will have a wider impact on society.

- Step 2: Propose and discuss the deep learning algorithms, datasets, models and other factors that you may have to consider for this application with the class.

- Step 3: Put together a flowchart in a Word document on how all the considered elements in Step 2 will come together. Clearly show the input and output, all the critical decision points, the processes that will take place in the middle and any possible errors that may occur in the process flow.

- Post your flowchart to the Module 3 discussion forum.

Learning Activity: Discussion Forum Post

- Using the deep learning resources, you have read in this Module, please identify at least two projects or research ideas that you would like to pursue as part of this unit.
- Describe each of these in one clear problem statement and write associated challenges or research questions for each statement. In this activity, you should demonstrate your ability to identify a research gap, propose a project to address the gap and communicate details in short written form. It should also help you to connect with likeminded people in the class.
- **Post your ideas to the Module 1 discussion forum.**
- **Please read other students' posts and provide feedback on their ideas. Most importantly, identify other students who are planning to work on projects similar to yours.**